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Professor Tamas Sziranyi
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Original title: “MEISCF: Multi-scale Edge Information Selection with Cross-scale Fusion for Small Object Detection in UAV Imagery”

Response to Reviewers

Dear Professor Sziranyi,

Thank you for the opportunity to revise this manuscript, and please pass our thanks to the six reviewers. Their central request was the same one, put in different ways: isolate the contribution of each component under matched conditions, and show that the reported gains are larger than the noise.

We ran those controls. They did not confirm the paper. Under a matched epoch budget, almost all of the accuracy we had attributed to our modules came from the training schedule, and the direction of that effect was the opposite of what we had claimed: single-resolution training at the deployment size beat our four-phase progressive schedule by 3.3 to 4.2 points mAP@50. A second seed reversed the sign of the MEIS and FRM difference. A factorial ablation showed that neither multi-scale dilation nor adaptive gating, the two ingredients that define MEIS, did better than a plain convolution with the same capacity. Resolved by object size, the modules improved no size range below 32 pixels.

We could not honestly repair the original manuscript, so we rewrote it as the controlled study the reviewers asked for. The revised paper reports what the experiments show, including the results that refute our earlier claims. Its new title is:

“What Drives Small Object Detection Accuracy in UAV Imagery? A Controlled Study of Training Resolution, Detection Stride, and Feature Modules”

We recognise that this is a larger change than a revision normally involves, and that the paper now argues against the method it was named after. We think it is the more useful paper, and the one the reviewers’ comments point to. If the Editor judges that it is better handled as a new submission, we will follow that advice.

Summary of the main changes

- **Matched comparisons.** Every comparison now varies one factor at a time at a matched epoch budget, with training and evaluation at the same input size. The schedule is crossed with the detection head in a 2×2 design (Table 5).

- **Statistical testing.** All differences are tested with a paired bootstrap and a paired permutation test over the 548 validation images, with Bonferroni correction across the ten classes. Two seeds are reported for the module comparison and for both schedules of the four-level detector (Sections 4.5, 5.4).
- **Component isolation.** A factorial ablation separates multi-scale dilation from adaptive gating and compares both against a plain convolution of equal capacity (Section 5.3.1, Table 7).
- **Accuracy by object size.** Results are resolved into COCO size ranges and into bands of network-input pixels tied to the detection strides (Section 5.5, Table 8, Fig. 8).
- **A predictive account.** Section 3 states a sampling argument and derives three predictions before the experiments, so the architectural results can be judged against a stated expectation rather than explained afterwards.
- **Cross-dataset evaluation.** UAVDT is now evaluated zero-shot with ignore regions honoured and sequence-level resampling, and the classes it cannot test are named (Section 5.6, Table 9).
- **Cost.** Throughput is measured where the GPU, not the host, is the limit, and reported next to the operation counts (Section 4.2, Table 2).
- **Claims removed.** The comparison against CF-YOLO under unmatched training, the “+17.6%” and “+10.2%” headline numbers, the claim of novelty for three-level fusion, and the unsupported activation-magnitude statistic have all been removed.

A note on cross-references. The manuscript was rewritten rather than edited, so line numbers of the original no longer locate anything. We therefore point to sections, tables and figures of the revised manuscript.

Reviewer #1

Introduction section could more clearly distinguish the genuine methodological contribution from the integration of previously established techniques. Several components . . . are based on well-known concepts. Please explicitly identify which aspects constitute the primary scientific contribution.

Response. We agree, and the controls we ran to answer this question showed that the components did not carry the result. The contribution is no longer a module. It is the controlled comparison itself: which of three design decisions — training resolution schedule, detection stride, and feature modules — actually moves small-object accuracy, measured one factor at a time with paired tests and two seeds, and resolved by object size. The Introduction now states this and lists three contributions: the matched-budget schedule-by-stride comparison, the controlled evaluation of the modules including the factorial ablation, and the sampling account with its size-resolved test (Section 1). We also state plainly that the modules we had proposed do not survive the controls.

The Related Work section . . . is largely descriptive. I recommend more critical discussion of previous approaches and a clearer explanation of how the proposed framework differs from recent lightweight YOLO variants, Transformer-based detectors, and other state-of-the-art methods.

Response. Related Work has been restructured around the four questions the study tests: feature modules for small aerial objects, resolution and sampling, resolution schedules during training, and variance and significance in detector evaluation (Sections 2.1–2.4). Each subsection now asks what evidence the cited work provides rather than what it proposes: how many seeds, whether the comparison was matched, and whether the reported gain exceeds the run-to-run spread. Section 2.4 is new and reviews what is known about seed variance in detection benchmarks, which is the standard against which we judge our own effects.

The largest improvement is achieved by the progressive multi-resolution training strategy, whereas the architectural modifications contribute only an additional improvement ... a more convincing evaluation should include comparisons with competing approaches, particularly CF-YOLO, trained using the same progressive training schedule.

Response. Your reading was correct, and the controlled experiment went further than we expected: the schedule dominates, and it dominates in the opposite direction. Single-phase training at 1280px beat the four-phase progressive schedule by 3.94 points mAP@50 with seed 0 and 4.16 with seed 42, with and without the stride-4 level (Table 4, Table 5; paired bootstrap 95% CI [4.1, 5.5] at 1536px, $p < 0.002$, permutation $p = 0.001$). At matched training compute the single-phase run was 3.87 points ahead (Section 5.1, Fig. 4).

We did not retrain CF-YOLO under our schedule. The comparison is no longer load-bearing, because the revised paper makes no claim of superiority over CF-YOLO or any other published method; that claim and its headline number have been removed. Section 5.7 and Table 10 now present published results with their protocols side by side — split, input size, initialization — precisely to show that they are not a ranking. Retraining a third-party detector under two schedules would be a worthwhile experiment, but its outcome would not change any conclusion we now draw, and we would rather not publish claims about another group’s method from a reimplementaion we cannot validate.

Adaptive gating mechanism within the MEIS module and the selected dilation rates are evaluated only as part of the complete framework.

Response. This is now a dedicated factorial experiment (Section 5.3.1, Table 7, Fig. 7). We crossed the dilation rates, (1, 2, 4) against (1, 1, 1), with the gate type, input-adaptive against a static learnable channel weight, holding the parameter count fixed. The corner with (1, 1, 1) dilations and a static gate is a plain residual block of depthwise-separable convolutions with the same capacity as MEIS. Neither ingredient beat that plain block on two seeds. Whatever MEIS contributes, it is not multi-scale dilation and not adaptive gating.

The methodological novelty of FRM also appears relatively limited ... its structure is essentially equivalent to the well-known CBAM module, while the primary difference lies only in its placement ... I could be taking into account deeper theoretical analysis explaining why this particular combination of modules should produce more discriminative feature representations.

Response. We accept the criticism and have stopped claiming otherwise. FRM is described as CBAM applied after fusion, with a citation to CBAM and no claim of structural novelty (Section 4.2). It is no longer counted as a separate contribution with its own percentage.

On the request for a theoretical account: rather than argue after the fact why the combination should work, Section 3 now states a sampling argument *before* the experiments and derives three testable predictions from it, with the object-size distribution of VisDrone measured in network-input pixels (Table 1, Figs. 1–2). The account predicts the stride results correctly and fails on the schedule result, and we say so (Section 6.1). This seemed to us more useful than a post-hoc justification of a combination that the measurements do not support.

Another aspect that deserves further attention is the interpretability ... I recommend present visualizations of feature maps, activation maps, or attention maps demonstrating that the network indeed focuses on edge-related structures.

Response. We have not added these, and we would like to explain why rather than simply decline. Such a visualization would be evidence for a claim the paper no longer makes. It can show that a branch responds to boundaries, but not that this response contributes to detection accuracy, and our measurements say it does not: MEIS improves no size range below 32 pixels on either seed, and a plain convolution of equal capacity does as well (Section 5.3, Table 8).

Publishing attention maps beside that result would invite the reader to trust the picture over the measurement. We report the quantitative decomposition instead, by object size and by component.

The cross-dataset evaluation on the UAVDT dataset considers only the dominant car category and results in a relatively modest improvement. Please ... evaluate the robustness ... under challenging imaging conditions, such as motion blur, varying illumination, partial occlusion, and image noise.

Response. The UAVDT evaluation has been rebuilt: zero-shot at 1280 px, official ignore regions honoured, vans merged into cars to match UAVDT’s label set, and confidence intervals from resampling whole sequences rather than frames, since frames within a sequence are not independent (Section 5.6, Table 9). The car ordering reproduces VisDrone — schedule +2.24 and +2.47 AP, stride +1.82, MEIS −0.81 — but no sequence-level interval excludes zero, and we now report that rather than the point estimates alone. We also state why trucks and buses cannot carry a conclusion: each has 62% of its boxes in a single sequence, so their AP moves by tens of points between resamples and the two classes rank the models in opposite orders.

We did not run a controlled corruption study. Doing it properly means calibrated blur, noise and illumination levels applied to a fixed image set, which is a separate experiment of its own size; and since the modules show no benefit in clean conditions on either seed, a robustness study would most likely measure the robustness of the baseline. We list it as future work in Section 6.4 rather than report a partial version of it.

I recommend extending the experimental comparison to include more recent lightweight Transformer-based object detectors as well as the latest YOLO variants specifically designed for aerial imagery.

Response. Section 2.1 now discusses a recent detection transformer for this task, UAV-DETR (Zhang et al., arXiv:2501.01855), alongside the aerial YOLO variants, and Section 2.4 was added to review how this literature handles run-to-run variance.

We did not, however, add new detector families to the experiments, for a reason that is central to the paper: comparing architectures without matched schedules, matched input sizes, matched epoch budgets and paired tests would reproduce exactly the problem the paper documents. A schedule change alone moves mAP@50 by about 4 points in our measurements, which is of the same order as the gains those detectors report over their own baselines — UAV-DETR reports +4.2 points AP@50 on VisDrone — so a table of detectors trained under their own recipes cannot separate architecture from protocol. Section 5.7 therefore reports published numbers with their protocols attached and explicitly declines to rank them, and Section 6.4 lists the narrow architectural scope — one detector family, one backbone scale — as a limitation of this study.

Reviewer #2

... the manuscript is clearly written, the technical contribution is well supported by the experimental results, and the proposed approach represents a valuable contribution to the literature.

Response. We thank the reviewer for the positive assessment, and we owe them a straightforward explanation of why the manuscript they recommended has changed so much. The controls requested by the other reviewers did not support the framework’s effectiveness. The experimental design the reviewer found comprehensive has been kept and extended, but its conclusions have reversed: the gains we reported came from the training schedule rather than from the modules, and the schedule itself works in the opposite direction to the one we claimed. We hope the revised paper remains useful to the community for the measurement practice it documents, even

though its findings are negative for the method.

Reviewer #3

The motivation is not clear. There are several methods combined together, what is the novelty of this manuscript and what are their functions?

Response. The manuscript no longer combines methods. It tests three design decisions separately and reports which one matters. The motivation, stated at the start of Section 1, is that gains of one to three points are routinely attributed to new modules in this literature while the training protocol is left uncontrolled and run-to-run variance is not reported; our own earlier results were an instance of this. The novelty is the controlled decomposition and its outcome, not a module.

Figs. 2 and 3 are confusing.

Response. The figure set has been rebuilt. The revised manuscript has eight figures: the sampling geometry that motivates the study, showing the same object under three stride and input-size settings (Fig. 1), the class distribution of the training set (Fig. 2), the detector architecture with the stride-4 path marked (Fig. 3), accuracy against training compute (Fig. 4), per-class differences (Fig. 5), a forest plot of every headline contrast with its confidence interval (Fig. 6), the factorial ablation (Fig. 7), and accuracy by object size (Fig. 8). Each is drawn to make one comparison, and the caption states what it shows.

The expression should be more concise. The contributions should be shortened.

Response. The manuscript has been rewritten throughout. The contribution list is now three short items in Section 1 rather than an extended enumeration, and the abstract is a single paragraph of results.

Please check the parts seem like being written by AI, for example, the former part of the abstract.

Response. The abstract has been written from scratch and now opens with the question the study asks and the design used to answer it, with no promotional framing. The manuscript as a whole was rewritten in plain declarative prose. Our use of AI-assisted tools is declared through the submission system, in line with the journal’s policy.

The references should be logically organized with enough analyses, and some outdated references should be deleted or replaced. Mohamed Meyer Kana Kone, et al. Applied Soft Computing, 2026, 203: 116069.

Response. Related Work is now organized by the four questions the study tests rather than by method family, and each entry is discussed in terms of the evidence it provides (Sections 2.1–2.4). The reference list was rebuilt for the new manuscript, superseded entries were removed, and the author lists and page ranges of the remaining entries were checked against the publishers’ records.

We were not able to locate the suggested Applied Soft Computing article from the details given, and the work we could find under that author name concerns plant disease identification and image watermarking rather than object detection in aerial imagery. We would gladly add it if the reviewer can supply the title, and we would equally welcome pointers to work that bears on the four questions above — particularly any aerial-detection study that reports run-to-run variance, since Section 2.4 had to draw most of its evidence from outside this literature.

Can quantum computing be helpful to solve this problem? NR Zhou, et al. ... Gong LH, et al. ...

Response. It is an interesting question, but outside what this manuscript can speak to. The study is an empirical comparison of training schedules, detection strides and feature modules for a convolutional detector on classical hardware, and none of its measurements bear on quantum methods. We have not added these citations, since we could not connect them to any claim in the paper, and a citation we cannot justify in the text would not serve the reader.

It is hard to follow since the theoretical foundation is not clear.

Response. Section 3 is new and supplies it. It sets out how the detection stride determines the sampling density of an object of a given size, measures the VisDrone size distribution in network-input pixels at each training resolution (Table 1), and derives three predictions: which size ranges a stride-4 level should help, which it should hurt, and what a module acting after downsampling can and cannot recover. The experiments then test those predictions, and Section 6.1 reports which held and which failed.

Why should we use YOLOv11 but not other new versions?

Response. Section 4.2 gives the reason: the study needs a detector whose neck and head can be modified in a controlled way and whose training pipeline is public and stable, so that the only difference between two runs is the factor under test. YOLOv11s provides this with COCO-pretrained weights at a scale that fits the repeated multi-seed training the design requires. The conclusions concern schedule, stride and module placement, which are not specific to the version; Section 6.4 states the single-family scope as a limitation.

More comparisons would be better to convince the audience. The results should be explained theoretically.

Response. The revised paper adds the comparisons that isolate causes rather than lengthen the ranking: the 2×2 schedule-by-stride design (Table 5), the factorial ablation of MEIS (Table 7), size-resolved accuracy for every contrast (Table 8), a second seed for four configurations, and a zero-shot cross-dataset evaluation (Table 9). Each result is set against the prediction of Section 3, and Section 6.1 separates what the sampling account explains — the whole pattern of the stride results — from what it does not — the schedule effect, for which we give the competing explanations and say which experiment would decide between them.

What is the cost of the performance improvement?

Response. Cost is now reported for both kinds of change. For training, we report pixel-epochs, a hardware-independent measure of the pixels a schedule processes (Section 4.3, Table 4), and accuracy against compute (Fig. 4). For inference, Table 2 reports parameters, operation counts at three input sizes and measured throughput at 1280 and 1536 px. The stride-4 level costs 74% more operations and 89% more time; MEIS with FRM costs 18% more operations but 85% more time, so its real cost is 1.6 times what its operation count implies, for no measurable accuracy benefit.

The advantages and disadvantages of this designed method were detailed, can it be realized with current technology?

Response. All models are ordinary convolutional detectors that train and run on a single consumer GPU; every number in the paper was produced on one RTX 3080, and Table 2 gives the measured throughput. The revised manuscript is explicit that the modules’ disadvantages outweigh their advantages: they cut throughput nearly in half at 1280 px and improve no size range below 32 pixels.

Reviewer #4

1. *The +1.6% gain from the MEIS module is a combined result of three design dimensions ... Is it the “edge extraction” itself that is effective? Or does the “adaptive gating” contribute the gain? Or is it simply that after resolution enhancement, any additional convolution can bring a +1.x% improvement?*

Response. The third possibility is the closest to what we measured. The factorial ablation (Section 5.3.1, Table 7) crosses dilation rates with gate type at a fixed parameter count, and its plain-convolution corner — (1, 1, 1) dilations, static gate — performs as well as full MEIS on two seeds. Moreover the +1.6% itself does not survive a second seed: measured against Baseline-P2 under matched conditions, MEIS with FRM changed mAP@50 by +0.58 points with seed 0 and −0.34 with seed 42 at 1536 px, with paired bootstrap intervals of [−0.15, 1.27] and [−1.00, 0.33] and permutation tests that agree (Section 5.3, Table 6, Fig. 6). Even on the favourable seed the interval excludes improvements larger than about 1.3 points. Split by object size, COCO small objects lost 0.8 points averaged over the two seeds, 95% CI [−1.5, −0.1] (Section 5.5).

2. *The authors claim to “simultaneously integrate P3/P4/P5,” but in fact only the P4 level achieves three-layer fusion ... Moreover, learnable scalar weights have already been widely used in CF-YOLO and EfficientDet.*

Response. Both observations are correct and the claim has been withdrawn. The neck is now described accurately, with its normalized learnable weighting attributed to the fast normalized fusion of BiFPN (Section 4.2), and no novelty is claimed for it. It is not part of the main experiments; the runs that used it are reported in Appendix A as earlier single-seed ablations, clearly labelled as such.

3. *The authors acknowledge that CF-YOLO did not use progressive training and that the comparison is an “upper bound.” However, this comparison is repeatedly emphasized as a +10.2% improvement in the Abstract, Introduction, and Conclusion, which can easily mislead readers.*

Response. Agreed, and it has been removed from the abstract, the introduction, the conclusion and everywhere else. The revised manuscript makes no comparative claim against CF-YOLO. Section 5.7 explains why published VisDrone numbers are not comparable across papers — split, input size, initialization and even the AP definition differ — and Table 10 lists protocols alongside numbers so that the rows cannot be read as a ranking.

4. *On UAVDT, evaluation is conducted only on the “car” class ... this also reveals that MEISCF’s cross-domain generalization capability for truly small objects has not been validated at all.*

Response. Correct, and this is now stated in the abstract rather than left to the discussion: UAVDT contains only vehicle classes, so cross-domain generalization for small non-vehicle objects remains untested. The rebuilt evaluation (Section 5.6) reports what the benchmark can and cannot support: cars reproduce the VisDrone ordering but no sequence-level interval excludes zero; trucks appear in six of eleven sequences with 62% of them in one, and buses in four sequences with 62% in one, so neither class can carry a conclusion. UAVDT’s objects are also larger, with 30.7% of instances below 32 pixels against 68.6% on the VisDrone validation set (Section 4.1), so the two benchmarks do not probe the same size regime.

5. *Table 1 and Table 3 appear to be redundant ... However, the reported metrics for the same methods are inconsistent across the two tables. Why?*

Response. The inconsistency was real and we apologize for it: those tables mixed runs from different configurations and input sizes. The revised manuscript uses one evaluation protocol for every number it reports — letterbox to the stated size, confidence 0.001, NMS IoU 0.5, at

most 600 detections per image (Section 4.4) — and any number produced under a different configuration is confined to Appendix A, which states the four ways that configuration differed. We also identified a second source of inconsistency that affects comparisons with the literature: Ultralytics releases differ in how the precision–recall curve is terminated, which changes mAP@50 by 0.5 to 0.7 points on identical detections. Section 4.4 states the definition we use.

6. Some experiments are conducted at 1024×1024 resolution, while others are at 1280×1280 . What is the rationale behind this choice? Why is the resolution not kept consistent across all experiments?

Response. There was no good rationale; the resolutions reflected the order in which the experiments had been run. In the revised manuscript every model is trained at its stated size and evaluated at 1280 and 1536 px, the same two sizes for all models. The 1024 px runs appear only in Appendix A, and one further 1024 px evaluation is used deliberately in Section 5.5 to test whether the schedule effect is a train–test scale mismatch, which it is not.

Reviewer #5

1. The authors acknowledge that the individual contribution of the per-input channel-wise gating ... was not isolated due to hardware constraints ... the authors should consider adding a lightweight theoretical justification or a highly scaled-down ablation ...

Response. We took this suggestion and were able to run the full ablation rather than a scaled-down one, using a shortened progressive schedule and two seeds (Section 5.3.1, Table 7, Fig. 7). Gate type and dilation rates are crossed at a fixed parameter count. Neither ingredient beat a plain convolution of equal capacity, so the +1.6% attributed to MEIS cannot be assigned to either of the mechanisms that define it — and under matched conditions with two seeds, the +1.6% itself does not replicate.

2. The cross-dataset evaluation on UAVDT focuses exclusively on the “car” category ... the authors should explicitly state in the abstract that “cross-domain generalization for small non-vehicle objects remains unvalidated.”

Response. Done, in those terms. The abstract now ends: “UAVDT has only vehicle classes, so small non-vehicle generalization remains untested.”

3. The +10.2% improvement over CF-YOLO is rightly noted as an upper bound ... A brief theoretical discussion ... estimating how much of the +12.8% curriculum learning gain might transfer to CF-YOLO would strengthen this comparative analysis.

Response. The premise of the estimate no longer holds: the curriculum did not produce a gain. Trained at a matched epoch budget, the progressive schedule was 3.3 to 4.2 points *worse* than single-resolution training at the deployment size, on both detection heads and both seeds (Tables 4 and 5). The comparison against CF-YOLO has been removed together with the claim it supported, so there is no transfer estimate to make.

4. To strengthen the literature review ... The authors are encouraged to review and integrate the following relevant frameworks: (a) UAV-DETR ... (b) PADrone: Pre-Flight Abnormalities Detection on Drones Using Deep RF Sensing ... (c) Multi-Sensor Fusion for UAV Classification Based on Feature Maps of Image and Radar Data.

Response. We have restructured Related Work around the four questions the study tests (Sections 2.1–2.4), and we have added a detection transformer to Section 2.1, which also answers Reviewer #1’s request for a recent Transformer-based detector.

One clarification on suggestion (a). Two different papers share the name UAV-DETR. The one at the identifier given, arXiv:2603.22841, is *UAV-DETR: DETR for Anti-Drone Target Detection*, which detects drones as targets — a different task from ours, where the camera is on the drone and the targets are small ground objects. The one that shares our task and our benchmark is Zhang et al., *UAV-DETR: Efficient End-to-End Object Detection for Unmanned Aerial Vehicle Imagery* (arXiv:2501.01855), which adds frequency-domain feature fusion, detail-preserving downsampling and semantic alignment to RT-DETR and reports +4.2 points AP@50 on VisDrone. We cite that one, and we use it to make a point the study depends on: its reported gain is of the same order as the effect we measure from the training schedule alone, which is why we compare factors rather than architectures.

On suggestions (b) and (c): this manuscript studies single-sensor optical detection of objects in aerial images, and its measurements concern detection stride, training resolution and module placement. The RF-sensing and image-radar fusion works address a different sensing modality and a different task, drone classification and pre-flight anomaly detection rather than detection of objects within an image, so we could not connect them to a claim in the paper without padding the review.

Reviewer #7

1. *The paper attributes the +17.6% total gain to a combination of training and architecture, but the +12.8% from progressive training is measured against a baseline that is not directly comparable. The architectural modules are only evaluated on top of progressive training, so their contribution is not isolated from the benefits of higher resolution inputs.*

Response. This diagnosis was right, and following it changed the paper’s conclusions. The revised design controls the training schedule explicitly: the progressive schedule and a single-phase schedule at 1280 px are crossed with the three-level and four-level heads at a matched budget of up to 350 epochs (Table 5). With the comparison made fair, the progressive schedule loses by 3.3 to 4.2 points mAP@50. The modules are evaluated against Baseline-P2 under an identical schedule and identical inputs, and the +17.6% figure has been withdrawn.

2. *The paper admits that FRM is “a placement choice rather than a structural novelty” ... However, the paper still counts FRM as a distinct module contributing +1.3%. This is a significant overstatement of novelty.*

Response. Agreed and corrected. FRM is presented as CBAM applied after fusion, with no claim of novelty and no separate percentage attributed to it (Section 4.2). Its effect is measured only as part of the MEIS + FRM contrast, which is not significant on either seed.

3. *...the +10.2% gap “represents an upper bound under unmatched training conditions.” This comparison is misleading and should not be presented as a fair state-of-the-art comparison.*

Response. Removed. See our response to Reviewer #4, point 3.

4. *Curriculum learning through progressive resolution increase is a well-established technique ... The paper presents this as a key innovation, but it is a standard training trick, not a methodological novelty.*

Response. We no longer present it as an innovation. Section 2.3 places progressive resolution training in its literature, and the revised paper treats it as a factor to be tested rather than a contribution to be claimed — a test it does not pass, since it underperforms single-resolution training at the deployment size under a matched epoch budget.

5. *The paper reports a +25% increase in GFLOPs ... and a +28% increase in parameters ... the model is no longer “lightweight” compared to the baseline it improves upon.*

Response. The cost is now reported without softening and measured more carefully than before (Table 2). Relative to YOLOv11s, the four-level head with MEIS and FRM carries 21% more parameters, 2.1 times the operations and 3.5 times the inference time at 1280 px. The word “lightweight” does not appear as a claim about our models.

We also found that the batch-size-1 timing convention used in the original submission understated these costs badly. With one image per batch, two of the three models were limited by how fast the host could submit work, not by the GPU: their forward pass took the same time at 1280 and at 1536 px. Read at face value, those numbers put the cost of the stride-4 level at 18% when a GPU-bound measurement puts it at 89%. Table 2 therefore reports both batch 1 and batch 8 throughput and marks the host-bound cells.

6. *The evaluation is performed only on the car category ... The +1.8% AP gain on large vehicles is not evidence of generalizable small-object detection.*

Response. Agreed. See our response to Reviewer #4, point 4. The revised Section 5.6 reports the car result with sequence-level confidence intervals that include zero, names the classes UAVDT cannot test, and the abstract states that small non-vehicle generalization remains untested.

7. *The paper reports a gap of 6x its uncertainty, but the uncertainty is derived from only three random seeds ... A proper significance test (e.g., a paired t-test across all validation images) is not performed.*

Response. Section 4.5 is new and describes the testing procedure: a paired bootstrap over the 548 validation images with 1000 resamples, a paired permutation test on the same pairing, Bonferroni correction across the ten classes, and sequence-level resampling for the video data of UAVDT. Every headline contrast is reported with its confidence interval, and Fig. 6 collects them in a forest plot.

Section 5.4 addresses the seed question directly and separates the two sources of uncertainty. For the module comparison the paired difference moved from +0.58 to −0.34 points between seeds, a standard deviation of 0.65 points, against an image-level bootstrap standard error of 0.35 — that is, the seed-to-seed spread was the larger of the two, and a single seed would have been misleading. The schedule effect is roughly ten times the seed spread and survives both tests. We recommend at least three seeds for effects below one point (Section 6.3), and we report that our own two seeds give only a rough estimate of that variance (Section 6.4).

8. *The paper states that “both models are evaluated at 1280×1280 resolution,” but the baseline was only trained at 640px. This gives MEISCF an unfair advantage ...*

Response. Correct, and this comparison no longer exists. Every model in the revised manuscript is trained at its stated input size and evaluated at 1280 and 1536 px, and each schedule’s final training resolution is 1280 px, so no model is asked to generalize to an unseen resolution in the headline comparisons (Sections 4.3–4.4). The only deliberate train–test size mismatch is the 1024px evaluation in Section 5.5, which is used to test a specific hypothesis about the schedule effect and is labelled as such.

9. *Table 2 shows “YOLOv11 Baseline + Progressive Training” achieving 50.3% mAP@50, but it is unclear if this model is trained for the same number of epochs (350) and with the same hyperparameters. A fair comparison would require a single-phase training at 1280px for the baseline.*

Response. That experiment is now in the paper and is one of its central results. YOLOv11s

was trained single-phase at 1280 px under the same budget of up to 350 epochs as its progressive counterpart, and it gained 3.28 points mAP@50 at 1280 px and 3.97 at 1536 px over the progressive schedule (Table 5; 95% CI [3.3, 4.6] at 1536 px, $p < 0.002$). Table 3 lists both schedules’ hyperparameters side by side, including the one difference we could not avoid: the three- and four-level detectors needed different physical batch sizes to fit in memory, which we disclose in Table 3 and list as a limitation in Section 6.4. We also note there that this single-phase run stopped early at epoch 345 under a patience of 200, with its best checkpoint at epoch 145.

10. Learnable weighted fusion of multi-scale features has been explored in prior work, including EfficientDet’s BiFPN and CF-YOLO. The paper’s claim that “simultaneous three-level fusion” is novel is weak.

Response. The claim has been withdrawn. See our response to Reviewer #4, point 2.

11. Dilated convolutions are typically used to increase receptive fields ... but edge detection typically benefits from fine-grained, small-kernel convolutions. The paper does not explain why dilation rates of 1, 2, and 4 are optimal for extracting edge information.

Response. We could not explain it, and the experiment shows there was nothing to explain: replacing (1, 2, 4) with (1, 1, 1) at a fixed parameter count changes nothing measurable (Section 5.3.1, Table 7). The revised manuscript makes no optimality claim for the dilation rates and reports this null result instead. The reviewer’s intuition about fine-grained kernels is consistent with what we measured, and with the sampling argument of Section 3: a module operating after downsampling cannot recover detail that the stride has already discarded.

12. The paper states that “ablation confirms balanced weighting outperforms stronger emphasis,” but it does not provide the ablation results.

Response. The sentence has been removed. The residual blend factor α sweep it referred to belongs to the earlier single-seed configuration and is now reported, with its actual numbers, in Appendix A (Table A.1), together with a statement of how that configuration differed from the main experiments.

13. The paper inconsistently reports performance at different resolutions ... A fair comparison would report FPS at the same resolution as the evaluation.

Response. Table 2 now reports throughput at 1280 and 1536 px, which are exactly the two resolutions at which every model is evaluated. As described in our response to point 5, we also changed how the measurement is made, because the batch-size-1 convention was measuring the host rather than the model.

14. Reference [26] is cited as “IEEE Access 10 (2022) 85744-85754,” but this appears to be a placeholder or a citation of a different paper ...

Response. The reviewer was right that the entry could not be verified. We were unable to locate it either, and it has been replaced by the paper that actually defines the MEIS block, Liu et al., *Intelligent and Converged Networks* 6(2), 2025, which is now the reference wherever MEIS-YOLO is discussed. The reference list was rebuilt for the revised manuscript and contains 30 entries, each cited at a specific claim in the text. We also checked the author lists and page ranges of the entries added in revision against the publishers’ records, and completed two that had been abbreviated.

15. The paper states that “68% of objects occupy fewer than 32×32 pixels” and that “small object features experience 4.2x weaker activation magnitudes,” but it does not show the analysis that led to these numbers.

Response. Both statements have been dealt with. The activation-magnitude claim was not supported by any analysis we could reconstruct, and it has been deleted. The size distribution is now derived rather than asserted. Section 3 gives a closed-form expression for the smallest object that can receive positive training samples at a given stride and input size, evaluates it in Table 1, and then measures the share of VisDrone objects falling below that threshold: at 640 px, 31.0% of validation instances are smaller than one stride-8 cell and 5.8% smaller than one stride-4 cell; at 1280 px the same objects give 5.8% and 0.2%. These numbers, not the earlier ones, are what the paper’s predictions rest on, and Section 6.1 checks the predictions against the measured results.

Closing

The reviewers asked us to isolate our components and to show that the gains exceeded the noise. We did both, and the answers went against the paper. We have reported them as we found them, removed the claims they do not support, and rebuilt the manuscript around the comparison that the evidence does support. We hope the result is useful, and we thank the reviewers for pushing the work to a place we would not have reached on our own.

Sincerely, on behalf of all authors,

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